(IPL)User Story 5 – IPL Processes Mission from UCRM via Adapter (Reverse Flow)

Story ID:US-IPL-P2P-005

Actor: IPL (Receiver System in reverse flow)

Goal: To receive and process missions (or equivalent reverse-flow messages) sent by UCRM via the Adapter.

Business Value: Enables bidirectional P2P communication, allowing UCRM to initiate actions or send information to IPL (missions), which IPL can then process in its own domain.

User Story

As an *IPL receiving system*, I want to receive and process missions sent by UCRM via the Adapter so that IPL can act on UCRM-initiated instructions or information according to UCRI 2.0 reverse flow.

Key Behaviours

- Expose an IPL interface/endpoint for Adapter to deliver missions.
- Validate incoming mission payload against agreed schema.
- Map mission data to IPL's internal structures and trigger appropriate business processes.

Acceptance Criteria

- Given Adapter forwards a mission from UCRM, when IPL receives the mission, then IPL validates the payload structure and mandatory fields (e.g., mission ID, type, correlation ID, timestamps) according to the agreed UCRI-based mapping.
- Given a valid mission payload, when IPL processes it, then IPL stores the mission and triggers the corresponding internal business logic (e.g., create/update internal mission, generate tasks, update status) according to IPL rules.
- Given an invalid mission payload (missing mandatory fields, malformed content), when IPL receives it, then IPL logs the validation error and provide the corresponding response.
- Given multiple missions are received in a short period, when IPL processes them, then IPL can handle them without data loss and, where required by business rules, preserves processing order for missions with dependencies.
- Given a duplicate mission is received (same mission ID or correlation ID), when IPL processes it, then IPL applies idempotent handling (e.g., ignore duplicate, log occurrence) to avoid double-processing.